

SHREYASH KALAL

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Education

Rutgers University-New Brunswick

Master of Science in Data Science

September 2023 – May 2025

New Brunswick, NJ

Professional Experience

Senior AI Engineer

May 2025 – Present

Capgemini – Cox Communications

United States

- Architected a multi-agent production platform automating Snowflake transformation SQL generation, validation, and dbt deployment across 30K tables and 5K daily executions, replacing manual transformation development.
- Designed the platform's generation layer using template-first Jinja-rendered SQL with RAG-based LLM fallback over an approved-SQL corpus, validated through SQLFluff linting, EXPLAIN, and row reconciliation.
- Pioneered the platform's human-in-the-loop review layer with Green/Yellow/Red risk-tiered triage and a continuous feedback loop promoting stable AI patterns into deterministic templates, expanding template coverage by 20%.
- Designed an Agentic AI Copilot for ETL maintenance with LLM-driven mapping normalization, a typed spec-diff engine, and deterministic serializers, cutting change-request turnaround from 1–3 days to 15–20 minutes across 56 artifacts.

Machine Learning Engineer

April 2022 – July 2023

Capgemini – Cox Communications

Bangalore, India

- Engineered a data validation framework on SageMaker using Robust PCA (recall 0.85), Kolmogorov-Smirnov drift tests, and Jensen-Shannon Divergence for categorical features, achieving 97% schema accuracy across S3 training datasets.
- Developed a time-series anomaly detection service combining STL, Prophet, and an RNN to monitor record-count drift (F1 0.83), wired to CloudWatch/SNS to trigger off-cycle retraining and restore a degraded churn model.
- Shipped a Redis-backed RESTful API for low-latency feature retrieval at online inference, delivering sub-100ms response times (15% faster) and cutting SQS-driven cycle time by 33% under peak load.

Senior Data Analyst

January 2022 – April 2022

Aurigo Software Technologies

Bangalore, India

- Built a BERT-based RFP scoring system precomputing embeddings and Levenshtein features for thousands of daily RFPs into a Parquet feature layer, fed to an LSTM via a SageMaker endpoint at 0.87 precision and sub-100ms latency.
- Designed a multi-label sentiment classifier on contract proposal responses using a CNN-to-Bi-LSTM architecture in PyTorch, tracked via W&B, achieving a Hamming Loss of 0.12 with reproducible training and evaluation harnesses.
- Delivered a Logistic Regression sales-win scoring model on Product Fit (from the RFP scoring service) and proposal-response features, achieving AUC 0.79, via a batch scoring pipeline feeding the CRM via automated CI/CD.

Data Analyst

September 2020 – December 2021

Aurigo Software Technologies

Bangalore, India

- Operationalized XGBoost models for customer churn propensity (recall 0.84) and lifetime value forecasting (R^2 0.88) on a monthly retraining cadence tracked in MLflow, with batch pipelines feeding CRM and BI consumers.
- Productionized a Gaussian Mixture segmentation model — with cluster count selected via BIC — scoring thousands of customers per batch run, persisting segment assignments to S3 for downstream marketing systems.
- Engineered the shared customer feature pipeline powering the churn, CLV, and segmentation models, reducing ML training data integration time by 23% with reusable, versioned features serving downstream models.

Projects

AI Usage Scoring Platform | *Python, FastAPI, Anthropic Claude, WebSocket, Judge0, SQLite, Fly.io*

- Devised event-sourced evaluation pipeline scoring AI usage across 3 dimensions via hybrid heuristic + LLM-as-judge, with Claude Sonnet 4.6 forced tool-use producing 150 evidence-cited binary judgments per session at \$1 cost.
- Deployed publicly with Judge0-sandboxed execution, per-session token caps, and per-IP rate limits; calibrated 9 judge questions to $\geq 90\%$ agreement on 180 hand-labeled fixtures with daily caps bounding worst-case spend.

Technical Skills

LLMs: OpenAI, Anthropic, Hugging Face, LangChain, LlamaIndex, Pinecone, Weaviate, FAISS, Embeddings, Fine-Tuning, LoRA, PEFT, LangSmith, RAGAS

Engineering: Python, TypeScript, FastAPI, Pydantic, Temporal, Prefect, Airflow, Docker, Kubernetes, CI/CD, REST APIs

ML: PyTorch, TensorFlow, scikit-learn, Transformers, Hugging Face Datasets, MLflow, Weights & Biases, A/B Testing

Cloud: AWS, GCP, Snowflake, dbt, PostgreSQL, Oracle, Redis, Parquet